



IBUPROFEN



The Small Molecule That Fights Pain

1 IDENTITY

IUPAC Name:

2-[4-(2-methylpropyl)phenyl]propanoic acid

Formula: C₁₃H₁₈O₂

Molar Mass: 206.28 g/mol

Class: NSAID (Nonsteroidal Anti-Inflammatory Drug)

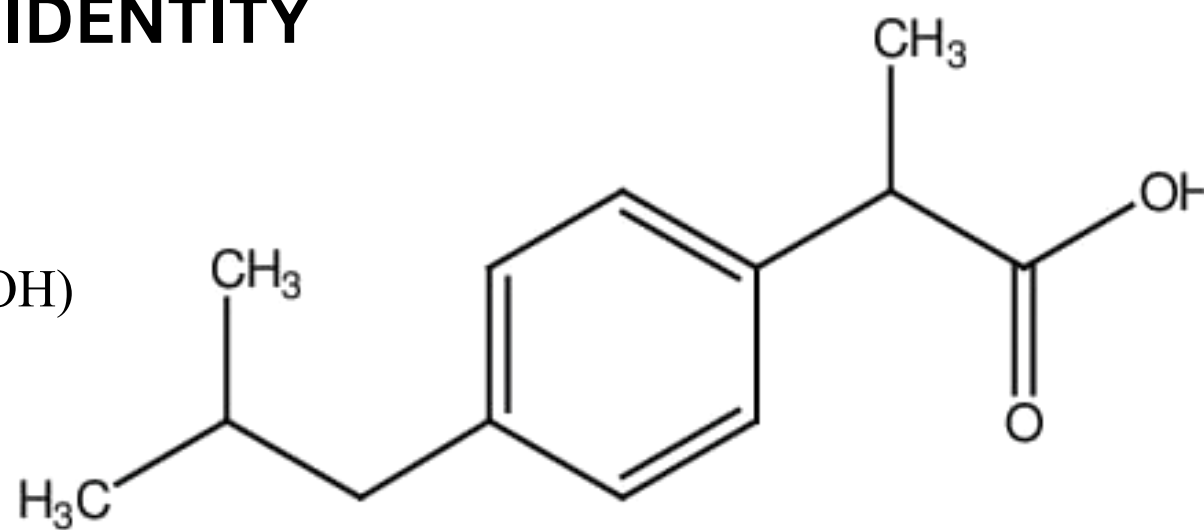
2 STRUCTURE & IDENTITY

Functional Group:

- Carboxylic acid (–COOH)
- Benzene ring
- Isobutyl group

Notable Feature:

- Chiral carbon (R/S)



COOL CHEMISTRY FACT

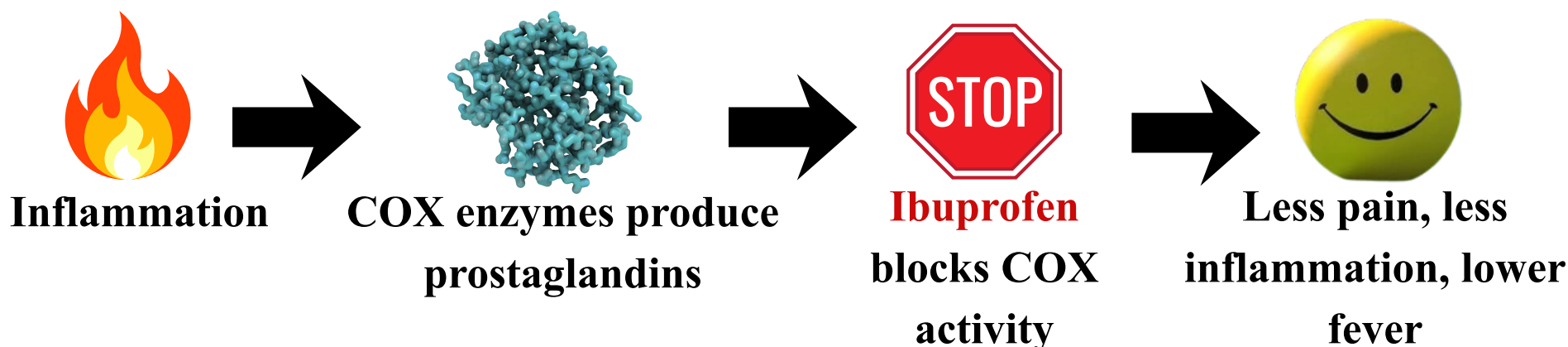
Ibuprofen has two mirror-image forms (R and S). The S-form is the main pain-relieving form.

3 IMPORTANT PROPERTIES

- White crystalline solid
- Melting point: ~75–77 °C
- Weak acid (pK_a ≈ 4.5)
- Poorly soluble in water
- More soluble in organic solvents
- Relatively lipophilic (LogP ≈ 3.5)

4 WHAT MAKES IT AMAZING?

INTERRUPTING PAIN SIGNALS!



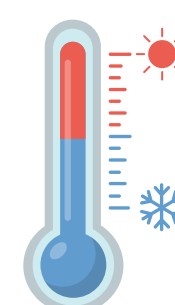
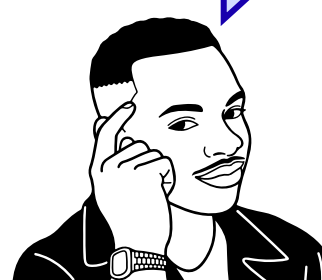
By inhibiting COX-1 and COX-2 enzymes, ibuprofen reduces prostaglandin production — the chemicals that contribute to pain and fever.

5 USES & VALUE

- Pain relief (headaches, toothaches, muscle and joint pain)
- Fever reduction
- Anti-inflammatory effect
- Treats arthritis and other inflammatory conditions

DID YOU KNOW?

One of the most widely used medicines in the world!

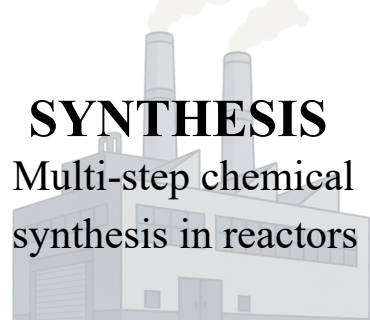


6 RISK & SAFETY

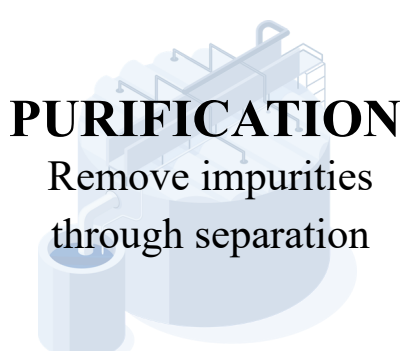
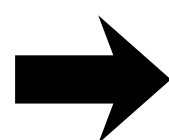


- May irritate the stomach or cause stomach bleeding
- May affect kidney function
- May increase cardiovascular risks with long-term use
- Use only as directed and avoid exceeding the recommended dose

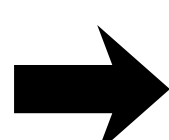
7 CHEMICAL ENGINEERING CONNECTION



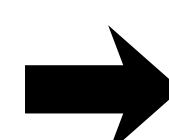
SYNTHESIS
Multi-step chemical synthesis in reactors



PURIFICATION
Remove impurities through separation



CRYSTALLIZATION
Obtain high-purity ibuprofen crystals



FORMULATION
Design tablets or capsules for safe, effective delivery

8 REFERENCE



- National Center for Biotechnology Information. (n.d.). PubChem compound summary for CID 3672, ibuprofen. PubChem.
- U.S. Food and Drug Administration. (n.d.). Organ-specific warnings for internal analgesic, antipyretic, and antirheumatic drug products for over-the-counter human use. FDA.